

Worksheet / Werkstuk 4.1- 4.5

Question 1 / Vraag 1

Given: / Gegee: $f(x) = \begin{cases} 4 - x^2 & \text{if } x \leq 1 \\ x - 1 & \text{if } x > 1 \end{cases}$

Is the function continuous on $\mathbb{R}$. Motivate your answer. / Is die funksie kontinu op $\mathbb{R}$.
Motiveer jou antwoord.

Question 2 / Vraag 2

Find the derivatives of the following functions / Bepaal die afgeleides van die volgende funksies.

1. $f(x) = 3^{x^2} e^{\sqrt{x}}$

6. $f(t) = 2 \cos(\ln t) - 4 \ln(1 + \cos 6t)$

11. $f(x) = \sinh(\sin x)$

2. $f(\theta) = \frac{\tan 2\theta}{1 + \sec(\theta^2)}$

7. $x^3 \sin y + xy^4 = y + e$

12. $f(x) = \cosh\left(\frac{1}{x}\right)$

3. $f(y) = \sqrt[3]{1 - \ln(1 + 2y)}$

8. $y = 2 \sin(x^3) + \sin^3(2x)$

13. $f(x) = \ln\left(\sin(2x^2) \sqrt{x^2 - 1}\right)$

4. $f(x) = \tan^{-1}(e^{2x})$

9. $f(\alpha) = 2^{3\alpha} + (3\alpha)^2 + 3^{2\alpha}$

14. $F(t) = \frac{t^2}{\sqrt{t^2 - 1}}$

5. $f(x) = \sin^{-1}(1 - 4t)$

10. $f(z) = z^{\cos z}$

15. $g(u) = \sin(u)^{\sin^{-1}u}$

Question 3 / Vraag 3

- Write down the domain and the critical points of the function $f(x) = \ln(4 + x^2)$. / Skryf neer die definisieversameling en die kritieke punte van die funksie $f(x) = \ln(4 + x^2)$
- Determine whether the function f satisfies the hypotheses of the mean value theorem on the given interval and determine the value(s) of c that satisfy the conclusion of the MVT. / Stel vas of die voorwaardes vir die middelwaardestelling vir funksie f bevredig is op die gegewe interval en bepaal die waarde(s) van c wat deur die MWS gewaarborg word.

$$f(x) = \ln(x^2 + 1), \quad x \in [0, 1]$$

- Determine the interval(s) on which the function $f(x) = x^3 e^{-2x}$ is decreasing. / Bepaal die interval(le) waarop die funksie $f(x) = x^3 e^{-2x}$ daal.
- Determine the interval(s) on which the function $f(x) = 4x^4 - \ln x$ is increasing. / Bepaal die interval(le) waarop die funksie $f(x) = 4x^4 - \ln x$ styg.
- Use the 1st derivative test to find the extreme values of / Gebruik die 1e afgeleide toets en vind die ekstreem waardes van

$$f(x) = x e^{-2x^2}$$

- Use the 2nd derivative test to find the extreme values of / Gebruik die 2e afgeleide toets en vind die ekstreem waardes van

$$f(x) = e^x \sin x, x \in [-2\pi, 2\pi]$$

- Use the closed interval method to find the absolute extremes of / Gebruik die geslote interval metode om die absolute ekstreem waardes te vind van

$$f(x) = e^{x^2 - 4x} \text{ on/op } [0, 3].$$

(You may use a calculator in this question / U mag 'n sakrekenaar in hierdie vraag gebruik)

8. Find the interval(s) on which the function $f(x) = x + \ln(x^2 + 1)$ is concave downward. / Vind die interval(le) waarop die funksie $f(x) = x + \ln(x^2 + 1)$ konkav na onder is.
9. Consider / Beskou

$$f(x) = \ln(1 - x^2)$$

- i Find the domain of f . / Bepaal die definisieversameling van f .
 - ii Find the vertical and horizontal asymptotes (if any) of f . / Bepaal die vertikale en horisontale asimptote (indien enige) van f .
10. Sketch a function with the following properties. / Skets 'n funksie met die volgende eienskappe.
- Domain / Definisieversameling $[-4, -2) \cup (-2, 2) \cup (2, 4]$
 - f is odd / f is onewe
 - $f'(x) < 0$ for all / vir alle $x \in [-4, -2) \cup (-2, 2) \cup (2, 4]$
 - $f''(x) < 0$ for all / vir alle $x \in (0, 2)$ and/en $f''(x) > 0$ for all / vir alle $x \in (2, 4)$
 - $\lim_{x \rightarrow 2^+} f(x) = \infty$ and / en $\lim_{x \rightarrow 2^-} f(x) = -\infty$

Question 4 / Vraag 4

Evaluate / Bepaal

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|---|---|---|--|
| 1. $\lim_{x \rightarrow 0^+} \ln x$ | 4. $\lim_{x \rightarrow \infty} \arctan x$ | 7. $\lim_{x \rightarrow \infty} \frac{x}{x-1}$ | 10. $\lim_{x \rightarrow 0^+} (1 + \frac{2}{x})^x$ |
| 2. $\lim_{x \rightarrow 2^+} \frac{x}{2-x}$ | 5. $\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 - x}}{x}$ | 8. $\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^3 - 4x}$ | 11. $\lim_{x \rightarrow \infty} (1 - \frac{2}{x})^{3x}$ |
| 3. $\lim_{x \rightarrow \infty} \frac{e^{4x}}{\ln x}$ | 6. $\lim_{x \rightarrow -\infty} \frac{e^{4x}}{\ln x }$ | 9. $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x}$ | 12. $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{\ln(x^2 + 1)}$ |